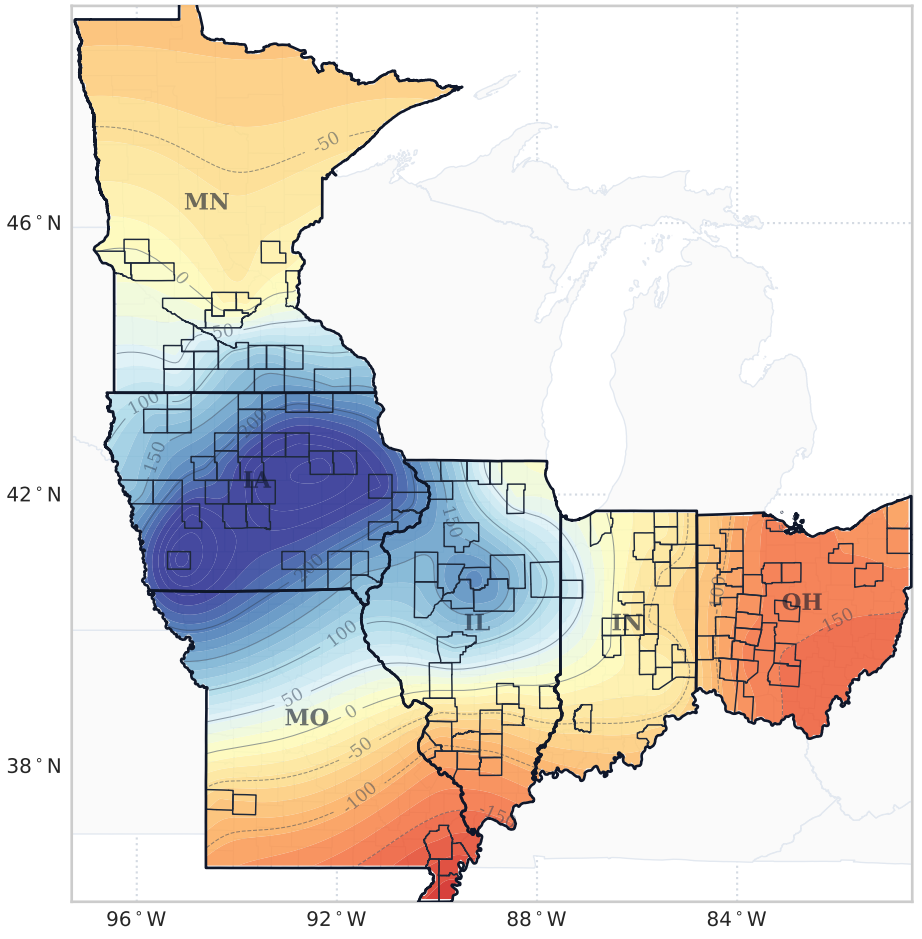
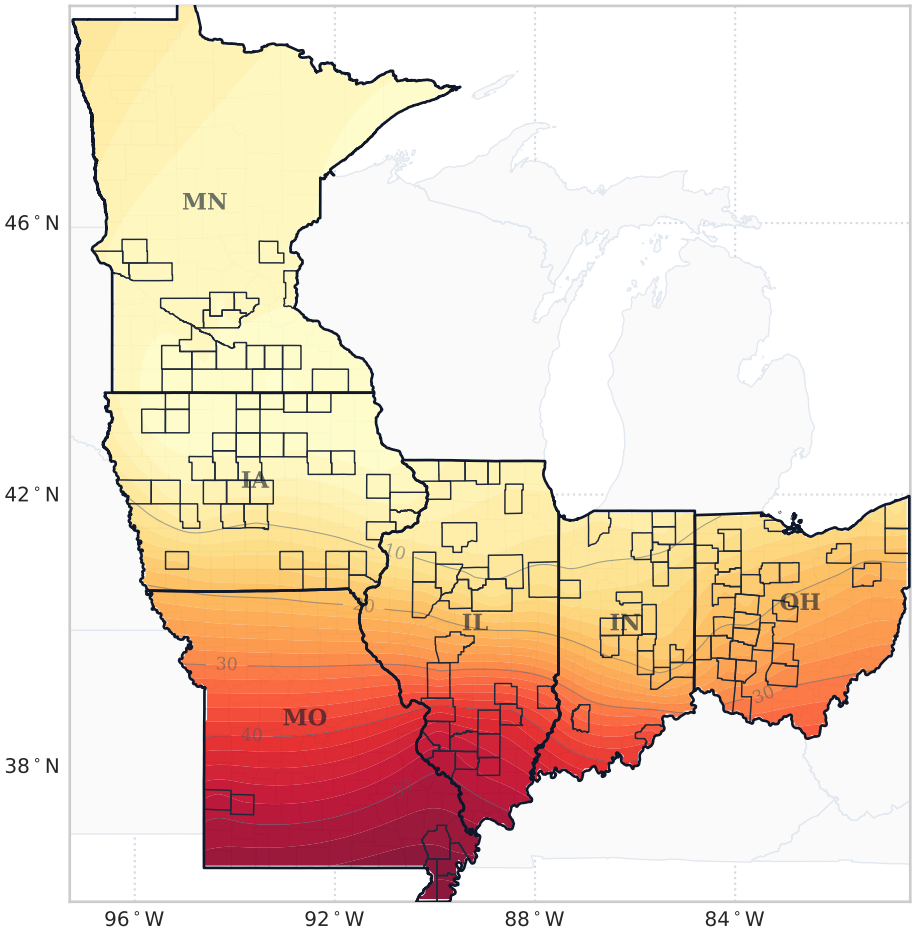
(a) 1993 Flood Shock: Net Water Balance ($P - ET_0$)



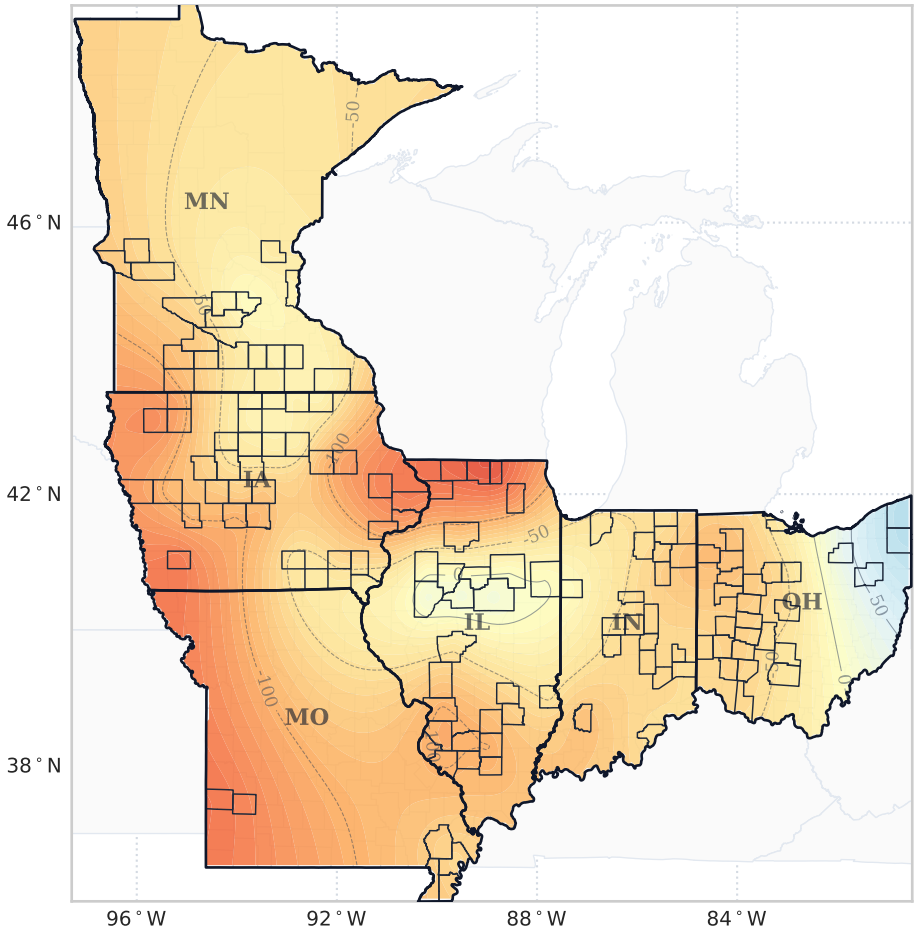
Yield Anomaly: -10.5% | Severe Pluvial Surplus West (>+200 mm) vs Deficit Southeast

(b) 1993 Flood Shock: Extreme Heat Days (HD_{30})



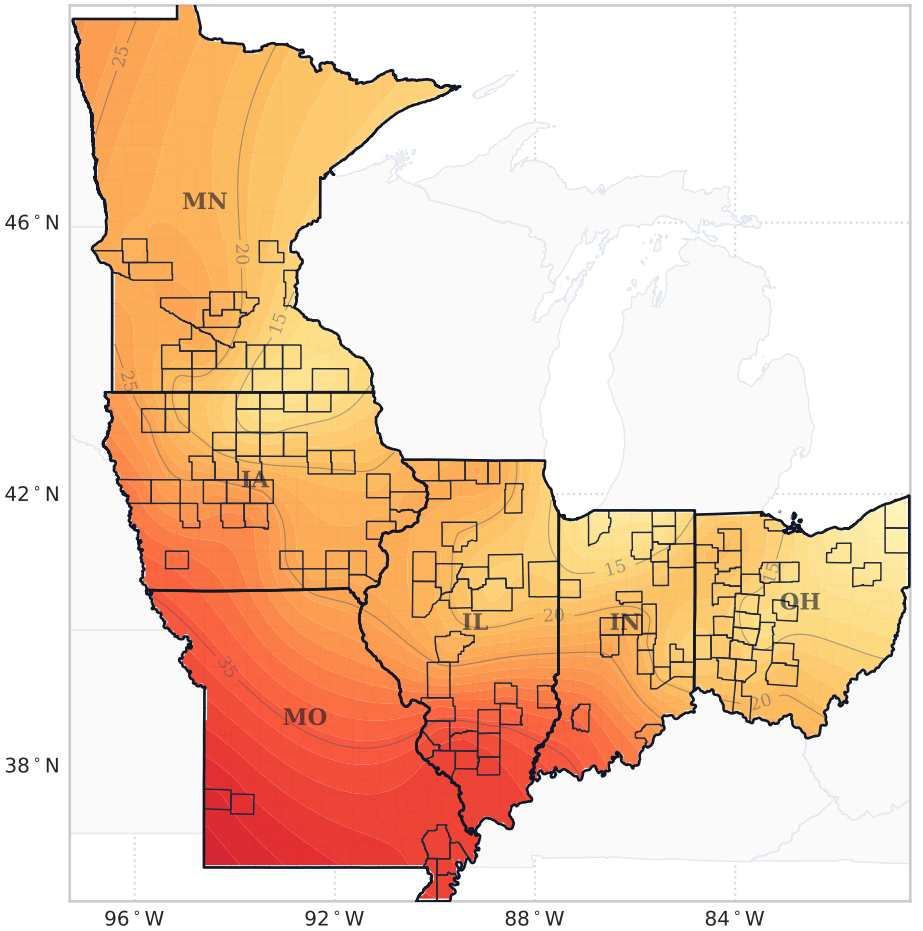
Yield Anomaly: -10.5% | Cold/Overcast North (<5 Days) vs Southern Heat (>50 Days)

(c) 2021 Modern Bumper: Net Water Balance ($P - ET_0$)



Yield Anomaly: +10.8% | Controlled, Favorable Hydrological Regime Across Midwest

(d) 2021 Modern Bumper: Extreme Heat Days (HD_{30})



Yield Anomaly: +10.8% | Balanced, Moderate Heat Accumulation Without Spikes



July--August Net Climatic Water Balance: $P - ET_0$ (mm) [Red = Deficit ← → Blue = Severe Pluvial Surplus]



July--August Extreme Heat Days: HD_{30} ($T_{max} \geq 30\,^{\circ}$C, days) [Yellow = Mild ← → Red = High Thermal Stress]